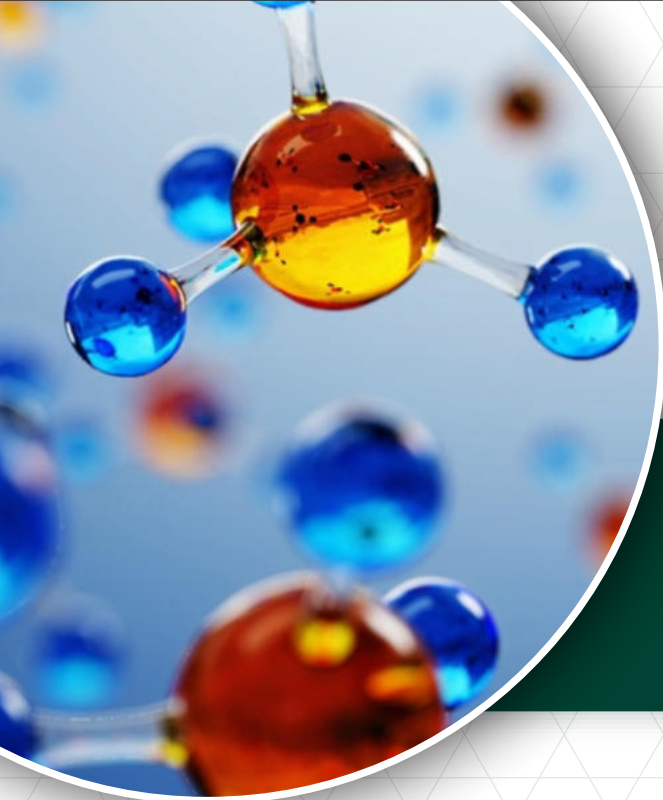




UNIT 08

INDUSTRIAL CHEMISTRY



SAPONIFICATION

Saponification is a chemical process that breaks down fats or oils into fatty acids and glycerol. It is commonly used to make soap. During saponification, a strong base like sodium hydroxide reacts with fats or oils, resulting in the formation of soap and glycerol. Soap molecules have properties that help remove dirt and grease. Saponification is also used in other processes like biodiesel production.

MATERIAL NEEDED FOR SOAP PREPARATION

1. ANIMAL FAT

Animal fats, such as tallow or lard, are derived from animals and serve as a source of fatty acids for soap production. They contribute to the cleansing and moisturizing properties of the soap.

2. PLANT OILS

Plant oils, such as olive oil, coconut oil, or palm oil, are obtained from various plants and are rich in fatty acids. These oils are commonly used in soap making as they provide different qualities and characteristics to the soap, including lathering and conditioning properties.

 Previous Year's Questions

What materials are required for preparation of soft drinks. [2025]

3. CAUSTIC SODA

Caustic soda, also known as sodium hydroxide (NaOH), is a strong base used in solid soap making. It reacts with the fats or oils during saponification, breaking them down into soap and glycerol.

4. POTASSIUM HYDROXIDE

Potassium hydroxide (KOH) is another strong base used in liquid soap making. It enables the saponification process, converting oils or fats into soap and glycerol. It is typically used for creating liquid or soft soap.

5. ADDITIVES

Additives are additional ingredients that can be incorporated into soap for various purposes. These can include colorants to give the soap a desired color, texture-enhancing ingredients for a smoother or exfoliating soap, and scents provided by essential oils or fragrance oils to add a pleasant aroma to the soap.

6. ABRASIVES

Abrasives such as silica, talc, or marble can be added to soap formulations to create exfoliating or scrubbing properties. These ingredients help remove dead skin cells and leave the skin feeling smoother.

SUGAR PREPARATION

Here are the steps for making sugar from sugar cane, as you mentioned:

1. HARVESTING AND DELIVERY

Sugar cane is harvested by cutting the stalks close to the ground when they are mature. The harvested sugar cane is then transported to the sugar mill for processing.

2. JUICE EXTRACTION

At the sugar mill, the sugar cane stalks are crushed to extract the sugary juice contained within. This is typically done using heavy rollers or mills that squeeze the juice out of the fibers.

3. CLARIFICATION

The extracted juice undergoes a clarification process to remove impurities. Lime (calcium oxide) or milk of lime (calcium hydroxide) is added to adjust the pH and aid in the coagulation and settling of impurities. This step helps remove non-sugar components and unwanted particles from the juice.

4. CONCENTRATION

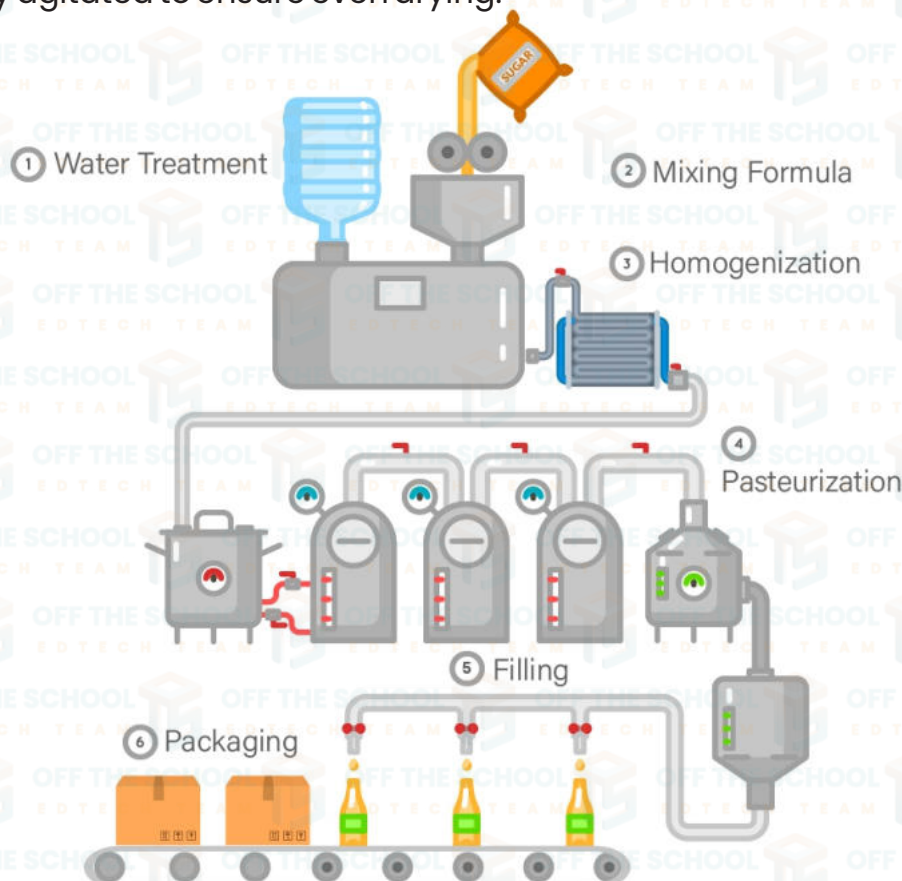
The clarified juice is then concentrated by evaporation. It goes through a series of evaporators where excess water is removed, and the sugar concentration increases. This process helps thicken the juice and concentrate the sugars.

5. CRYSTALLIZATION

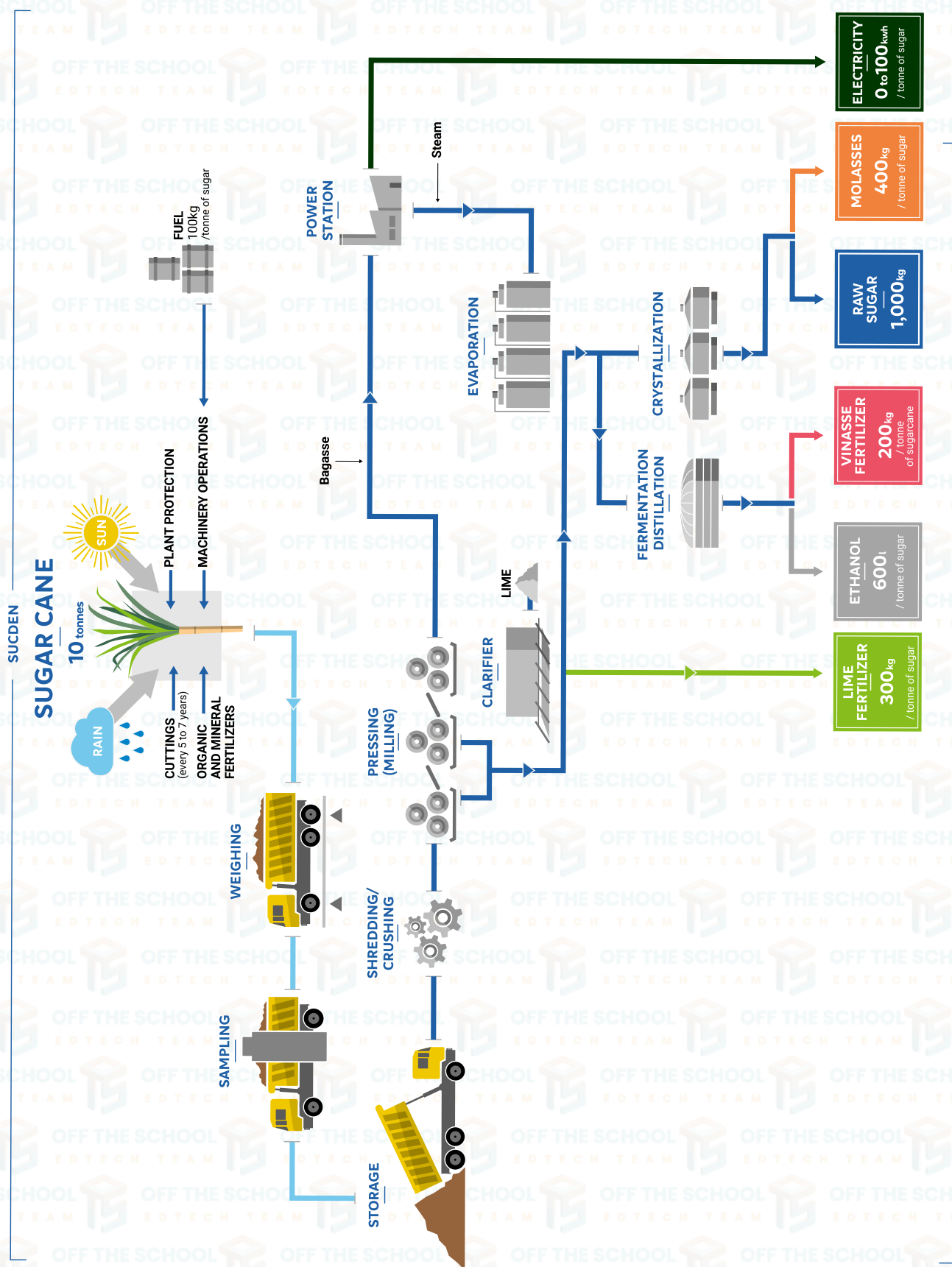
Once the juice is concentrated to the desired sugar concentration, it is further processed to induce sugar crystallization. This is achieved by cooling the concentrated juice and either adding seed crystals or using specialized equipment designed to promote crystal formation. As the juice cools, sugar molecules come together and form crystals.

6. CRYSTAL SEPARATION AND DRYING

The sugar crystals are separated from the remaining liquid, known as molasses. This is typically done using centrifuges, which spin the mixture at high speeds. The centrifugal force separates the sugar crystals from the molasses. After separation, the sugar crystals are dried to remove any remaining moisture. This is done using hot air, and the crystals are continuously agitated to ensure even drying.



SUGAR CANE PROCESSING



PETROLEUM

Petroleum, commonly known as crude oil, is a naturally occurring fossil fuel derived from ancient organic matter. It is a complex mixture of hydrocarbons, with varying compositions and properties. The formation of petroleum and natural gas involves the following processes:

Fractional Distillation

Fractional distillation is a process used to separate mixtures of liquids. It relies on the substances in the mixture having different boiling points. When heated, the mixture releases some of the substances as a vapor, which cools and condenses back into a liquid state. The different substances do this at different temperatures.

Fraction distillation of crude oil

Depending on the size of their molecules, the hydrocarbons in oil are solid, liquid, or gas at room temperature. Fractional distillation separates oil into “fractions.” A “fraction” is a mixture of molecules with similar boiling points.



Key Facts

- ✓ Fractional distillation separates crude oil into useful mixtures.
- ✓ Fractions are called “fractions” because they are only part of the original crude oil.
- ✓ Hydrocarbons in a fraction have a similar boiling point and number of carbon atoms.
- ✓ Going up the column, fractions become less viscous, easier to ignite, and have lower boiling points.

Substances that are gases at room temperature collect at the top of the column.

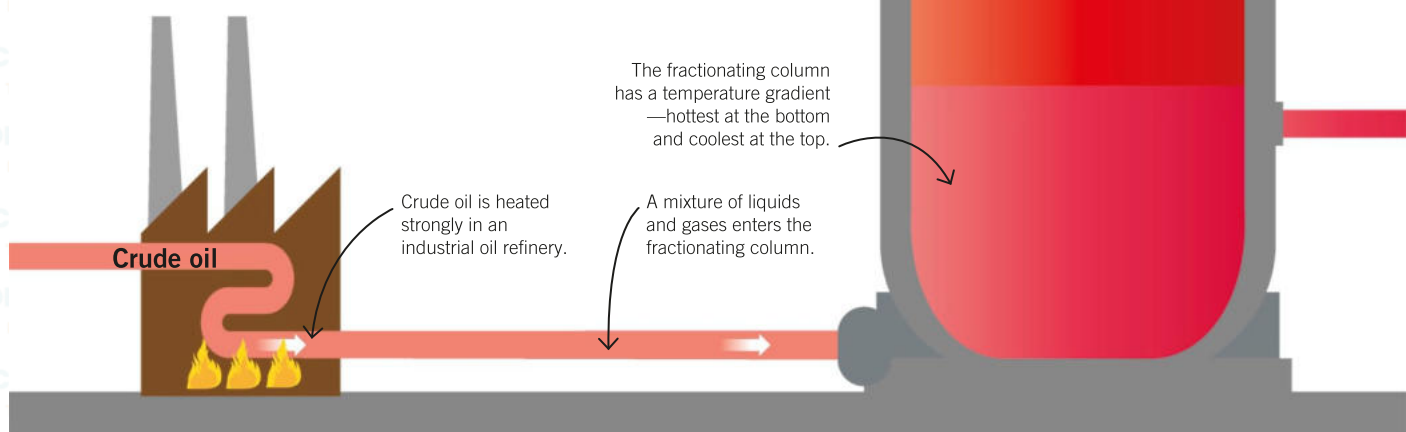
The higher up the column, the runnier and easier to ignite the fractions become.

The vapor cools as it rises, with each substance condensing back into liquid at a different temperature.

The fractionating column has a temperature gradient—hottest at the bottom and coolest at the top.

Crude oil is heated strongly in an industrial oil refinery.

A mixture of liquids and gases enters the fractionating column.



1–4 carbon atoms Highly flammable gases 		Refinery gases Heating and cooking
4–12 carbon atoms Highly flammable liquids with very low viscosity 		Gasoline Fuel for cars
7–14 carbon atoms Very flammable liquids with low viscosity 		Naphtha Feedstock for the petrochemical industry
11–15 carbon atoms Flammable liquids with low viscosity 		Kerosene Fuel for aircraft
14–19 carbon atoms Flammable but viscous liquids 		Diesel oil Fuel for some trains and cars
18–30 carbon atoms Very viscous liquids, difficult to ignite 		Fuel oil Fuel for large ships and some power stations
30+ carbon atoms Solids at room temperature 		Asphalt Waterproofing roofs and surfacing roads

ORIGIN OF PHARMA

medicinal purposes. Ancient Egyptians, Chinese, and Greeks documented the use of various herbs and plants for treating ailments. The development of pharmaceutical science progressed over time, with advancements in chemistry and the discovery of active compounds leading to the production of modern drugs. Today, the pharmaceutical industry plays a vital role in researching, developing, and manufacturing medications for the prevention, treatment, and cure of diseases.

IMPORTANCE OF PHARMACEUTICAL INDUSTRY

The origin of the pharmaceutical industry holds great importance due to its numerous benefits:

1. INCREASED LIFE EXPECTANCY

Pharmaceutical treatments have significantly contributed to increasing life expectancy by providing effective medications for various diseases and conditions.

2. DISEASE ERADICATION AND ELIMINATION

The industry strives to eradicate and eliminate diseases through the development of vaccines and innovative therapies, leading to improved public health outcomes.

3. REDUCED PAIN AND SUFFERING

Pharmaceutical advancements have resulted in the availability of pain-relieving medications, providing relief and improved quality of life for individuals suffering from chronic pain or serious illnesses.

4. COST SAVINGS FROM VACCINES

Vaccines play a crucial role in preventing infectious diseases, leading to reduced healthcare costs associated with treatments, hospitalizations, and long-term care.

5. SHORTER HOSPITAL STAYS

Effective pharmaceutical interventions have led to shorter hospital stays, as certain conditions can be treated more efficiently and effectively, reducing patient recovery time.

6. JOB CREATION

The pharmaceutical industry employs millions of people worldwide, contributing to job creation, economic growth, and stability in various regions.

7. ECONOMIC BOOST

Pharmaceutical companies contribute to the global economy through investments in research and development, production, sales, and exports, stimulating economic growth and innovation.